

Chapter 87 — Surgery of the Scrotum and Seminal Vesicles

SURGICAL ANATOMY OF THE SCROTUM

Scrotal Wall

- **Anatomy governs technique:** safe entry, recognition of planes at debridement, multilayer closure.
- **Skin is rugated** because it is attached to the underlying dartos muscle; it is pigmented and hair-bearing.
- **Thin and highly elastic**, so the scrotum changes size with temperature to support spermatogenesis.
- **Loose tissue is forgiving** to handle, but the same distensibility gives **poor tamponade**: bleeding is not contained, haematomas enlarge and may need re-intervention. **Meticulous haemostasis is therefore mandatory.**

Scrotal Contents

- **Held in place** by the spermatic cord and the gubernacular attachments.
- **Coverings, testis outwards:** tunica albuginea → tunica vaginalis → internal spermatic fascia → cremaster muscle → external spermatic fascia → dartos → skin.
- **The dartos fascia is a barrier** to spread of necrotising fasciitis to the deeper scrotal structures.
- **Epididymis:** arises from the efferent ducts of the rete testis in the mediastinum testis and lies **posterolaterally** — caput superior, corpus along the long axis, cauda at the lower pole.
- **Vas deferens:** continuous with the cauda → spermatic cord → pelvis → ampulla → joins the seminal vesicle (SV) → proximal ejaculatory duct.
- **Two vestigial remnants:** the appendix testis on the upper pole of the testis and the appendix epididymis on the head of the epididymis.

Figures 87.1 and 87.2 — anatomical barriers to the spread of infection (modified from Kavoussi and Costabile, *Practical Urology*, 2011).

- **Coronal and sagittal views** label the superficial fascial system — scrotal dartos, Colles fascia of the perineum, Scarpa fascia and penile Buck fascia — and, deep to the dartos, the spermatic fasciae enclosing the testis.

Vasculature

ARTERIAL SUPPLY OF THE SCROTUM, TESTIS AND EPIDIDYMISS

Territory	Artery	Parent vessel
Scrotum, anterior	Superficial and deep external pudendal arteries	Femoral artery
Scrotum, posterior	Posterior scrotal artery	Perineal artery (internal pudendal, anterior division of internal iliac)
Testis, primary	Testicular artery	Aorta
Testis, secondary	Artery to the ductus deferens (deferential artery)	Superior vesical artery ← internal iliac
Testis, secondary	Cremasteric artery	Inferior epigastric artery ← external iliac
Epididymis, dominant	Superior epididymal artery	Testicular artery

Territory	Artery	Parent vessel
Epididymis, minor	Inferior epididymal artery	Deferential artery and distal testicular branches

- **Venous drainage** follows the arteries: external pudendal and posterior scrotal veins → great saphenous and internal iliac veins.
- **What the table tests:** the testis has **three sources from three different parent trunks** (aorta, internal iliac, external iliac).
- **Anastomoses:** cranially, fine testicular–deferential anastomoses in the cord about 10 cm above the testis; caudally, testicular–deferential–cremasteric anastomoses at the cauda epididymis.
- **Consequence:** the testis survives injury or ligation of one or two arteries — hence the testicular artery can be ligated for extra length in the **Fowler–Stephens orchidopexy** for undescended testis, with low risk.
- **Superior epididymal artery** carries most epididymal flow, yet can be sacrificed to mobilise the testis in vasectomy reversal because it anastomoses with the inferior epididymal artery.
- **Transdermal absorption:** thin, vascular scrotal skin absorbs up to about 40 times more than other skin — exploited by scrotal testosterone patches, since withdrawn because frequent scrotal hair clipping was impractical.
- **Scrotal skin is rich in 5 α -reductase**, so scrotal testosterone produced high dihydrotestosterone (DHT) levels.

Innervation

AUTONOMIC INNERVATION OF THE TESTIS — THREE CONVERGING GROUPS

Group	Origin	Course
Superior spermatic nerves	Renal and mesenteric plexuses	Follow the testicular artery
Middle spermatic nerves	Superior hypogastric plexus	Run beside the vas as it enters the internal ring
Inferior spermatic nerves	Inferior hypogastric plexus	Fuse with the middle group at the prostatovesical junction, run along the vas in the cord, penetrate epididymis and testis

- **Afferent autonomic fibres** carry sensation and also take part in **endocrine control** of the testis: in rats, noradrenaline applied to inferior mesenteric plexus ganglia raised serum testosterone, whereas superior mesenteric ganglia had no effect.

SOMATIC INNERVATION OF THE SCROTUM

Nerve	Position	Territory	Effect of injury
Genital branch of genitofemoral nerve	Inferolateral aspect of spermatic cord	Anterolateral scrotal skin, tunica vaginalis, cremaster	Pain in its distribution; loss of cremasteric reflex
Ilioinguinal nerve	On the anterior surface of the cremaster	Anterior scrotal skin	—
Posterior scrotal nerves	Branches of the pudendal nerve	Posterior scrotal skin	—

- **Clinical use:** knowing these courses allows diagnosis of neuropathic scrotal pain and planning of scrotal denervation procedures.

Concept — The scrotal wall is supplied anteriorly from the groin (external pudendal vessels, ilioinguinal and genitofemoral nerves) and posteriorly from the perineum (pudendal system), whereas the testis has its own triply redundant supply from aorta, internal iliac and external iliac; that redundancy is what makes arterial ligation for mobilisation safe and cord denervation possible without devascularisation.

PREOPERATIVE CONSIDERATIONS

- **Hair removal** improves exposure, reduces surgical site infection, and eases suturing and dressing.
- **Razor rather than clippers for the scrotum:** a comparative study found less skin trauma and a better shave with razors, although hospitals generally favour clippers → **disposable razor on the day of surgery.**
- **Antisepsis:** iodine solutions or chlorhexidine gluconate. A Cochrane review favoured chlorhexidine scrub over alcohol-based povidone-iodine, but on small, poor-quality trials → **choose on cost and allergy profile.**
- **Antibiotic prophylaxis — American Urological Association (AUA) Best Practice Statement:** for genital surgery, a **single dose of intravenous cephalosporin or clindamycin only in patients with risk factors** — advanced age, anatomical urinary tract abnormality, poor nutrition, smoking, chronic corticosteroid use, immunodeficiency, externalised catheters, colonised endogenous or exogenous material, distant coexistent infection, prolonged hospitalisation.
- **Anaesthesia:** local, regional, spinal or general; **subcutaneous infiltration plus spermatic cord block** allows many scrotal procedures in the office.
- **Preincisional 0.5% lidocaine without adrenaline** is recommended in theatre or office to control postprocedural pain.

Surgical Approaches to Scrotal Contents

Access Into the Scrotum

GOVERNING CONTRAST — APPROACH BY AETIOLOGY

Benign disease	Possible malignancy
Scrotal incision	Inguinal incision
Direct opening of the tunica vaginalis	Cord isolated and clamped with non-crushing clamps before the testis is handled, to prevent tumour seeding → frozen section
—	Frozen section negative → testis and cord spared, operation completed

- **Scrotal hypothermia** may be used once the cord is clamped; however, animal models show worse testicular function and histology after cold than after warm ischaemia.
- **Incision:** transverse or median raphe. **The median raphe is optimal** because the superficial dartos forms an avascular median plane between the hemiscrota.
- **Common to all techniques:** secure haemostasis as the tunica vaginalis is bluntly freed from the scrotal wall, and avoid injuring testis, epididymis and cord when incising the tunica.
- **Obstructive azoospermia** can follow epididymal or vasal injury during hydrocelectomy — take care when opening and excising the sac.
- **Closure:** **always close the dartos** (running or interrupted) — **most haematomas arise from inadequate dartos closure.**

Concept — *Approach follows pathology: the inguinal route controls the cord before a possible tumour is touched, and the scrotal route is used only once malignancy is excluded; because the scrotum cannot tamponade, closure of the dartos is the haemostatic step that prevents the commonest complication.*

SURGERIES OF THE SCROTUM

Scrotal Wall

Cyst/Tumour Excision

- **Sebaceous and epidermal inclusion cysts** are managed conservatively or excised if not infected; **standard ellipsoid excision** gives excellent outcomes and cosmesis.
- **Infected cyst:** antibiotics or incision and drainage (I&D) with systemic therapy, depending on size and extent — possible in clinic or emergency department with minimal analgesia.
- **Excise once infection resolves**, because recurrence is relatively high.

Partial/Total Scrotoectomy

- **Partial scrotoectomy is uncommon;** the main indication is **Fournier gangrene (FG)**.
- **FG mainstay:** intravenous fluids, broad-spectrum antibiotics and **urgent surgical debridement** — gangrene can spread **2–3 cm per hour**, and delay significantly increases mortality.
- **Mortality about 20%** despite aggressive treatment, unchanged over 25 years.
- **Repeat debridement every 24–48 hours** until all visibly necrotic tissue is removed (Fig. 87.3A); on average up to four procedures.
- **Multidisciplinary care:** trauma, general and colorectal surgery, intensive care, infectious disease.
- **Anorectal extension** may require faecal diversion by colostomy — an essential planning consideration.
- **Scrotal violation in non-seminomatous germ cell tumour (NSGCT):** after trans-scrotal aspiration, biopsy, exploration or orchidectomy, **partial scrotoectomy** is required to prevent contamination and local recurrence; a systematic review found residual primary tumour in a minority of scar excisions.
- **Other indications for extensive excision:** hidradenitis suppurativa, post-radiation lymphoedema, penoscrotal Paget disease, primary lymphangitis; also cosmetic reduction of scrotal descent.
- **Total scrotoectomy** is rarer: extensive FG, malignancy, extensive trauma.
- **If more than 50–60% of scrotal skin is removed**, no immediate reconstruction: the testes go into **medial thigh pockets** with loose closure until reconstruction, or are wrapped in saline gauze or included in the vacuum-assisted closure (VAC) dressing.

Scrotal Reconstruction

- **Depends on** the extent of the defect and availability of suitable local tissue.
- **Minor defects:** wet-to-dry dressings and healing by secondary intention.
- **Significant defects (loss of more than half the scrotal skin): negative-pressure wound therapy (VAC)** — removes bacteria and oedema, improves angiogenesis, promotes granulation → faster recovery, smaller wound, better graft take.

FIGURE 87.4 ALGORITHM FOR THE MANAGEMENT OF SCROTAL RECONSTRUCTION FOLLOWING SCROTAL DEBRIDEMENT

Defect	Condition	Management
<50% of scrotum	Closes without tension	Primary closure
<50% of scrotum	Tension	Local scrotal advancement flap or secondary intention
>50% and extending beyond scrotum	—	Split-thickness skin graft (STSG) ± tissue adhesive, or flap reconstruction

Modified from Karian, Chung and Lee, *Eplasty*, 2015 (systematic review).

- **Less than half the scrotum:** secondary intention, helped by loose approximation with non-absorbable monofilament and VAC.
- **More than half:** local advancement flaps from the remaining scrotum or thigh; **remaining scrotal skin and dartos can be undermined and stretched** to cover large defects.
- **Delayed primary closure** with nylon horizontal mattress sutures once infection resolves: complications occur (dehiscence, flap necrosis, bleeding), but hospital stay is shorter and cost probably lower than secondary intention (retrospective review).
- **No local tissue → fasciocutaneous thigh flaps** (superomedial or anterolateral), with good cosmesis.
- **Meshed STSG** is usable but **inferior to thigh flaps** because of contracture.

Figure 87.3 — (A) aggressive debridement of FG; (B, C) wide excision of perineal hidradenitis suppurativa, before and after; (D, E) excision and STSG for penoscrotal lymphoedema, before and after (A from Kavoussi and Costabile, 2011).

Concept — Scrotal elasticity makes the scrotum its own best donor: reconstruction climbs a ladder from secondary intention, to stretched local scrotal skin, to thigh flaps, with grafts last because contracture defeats a tissue whose function depends on laxity.

Testis

Hydrocelectomy

- **Hydrocele** = fluid between the parietal and visceral layers of the tunica vaginalis.

GOVERNING CONTRAST — CONGENITAL VERSUS ACQUIRED HYDROCELE

Congenital	Acquired
Communicating — peritoneal fluid through a patent processus vaginalis	Imbalance of production and reabsorption within the tunica vaginalis
Commoner in preterm neonates	Among the commonest benign scrotal conditions in men over 40
Processus closes typically by the second year; about 80% resolve spontaneously by 18 months	A little peritesticular fluid is usual even in asymptomatic men on ultrasound

- **Fluid may be** peritoneal fluid, lymph, pus, blood, bile, cerebrospinal fluid or urine.
- **Presentation:** painless swelling, which may extend along the cord into the inguinal canal.
- **Treat only for** discomfort, cosmesis or underlying malignancy; **urgent treatment for pyocele** with signs of infection.
- **High-resolution B-mode and colour Doppler ultrasonography** distinguishes the causes of fluid, can reveal intratesticular malignancy, and guides the approach.
- **Excisional techniques give the lowest recurrence** and are reserved for chronic, large, multiloculated, thick-walled hydroceles; **plication suits thin-walled, smaller hydroceles.**

COMPARISON OF HYDROCELECTOMY TECHNIQUES

Technique	Principle	Key steps	Suited to / feature
Lord plication	No excision	Open sac anteriorly → deliver testis through aperture → plicate sac with interrupted, radially placed circumferential chromic sutures (Fig. 87.5)	Thin-walled, smaller; lowest haematoma risk, no drain needed

Technique	Principle	Key steps	Suited to / feature
Jaboulay (Winkelman)	Excise and evert	Resect part of parietal tunica vaginalis → evert remainder behind testis → approximate edges without compressing the cord ; or excise excess, cauterise, oversew cut edges with 3-0 chromic	Thin, floppy sacs (Fig. 87.6)
Bottleneck	Excise, leave cuff	Trim all but a 2-cm circumferential cuff around testis and cord → tack edges together; sac left open	Excisional
Window	Fenestrate	4-cm anterior scrotal incision → bluntly clear parietal layer → stab drainage → 2.5 × 2.5-cm cruciate window in parietal layer → evert flaps, stitch with 2-0 chromic	Less commonly used
Simple excision	Excise	Excise sac, oversew edges (Fig. 87.7)	Thick-walled sacs

- **Pattern across the rows:** the more tunica excised, the lower the recurrence but the larger the raw surface and haematoma risk; plication trades some recurrence risk for a dry field.

Aspiration and Sclerotherapy

- **History:** William of Saliceto instilled sugar and ginger in the 13th century; sclerotherapy re-emerged in 1975 as a cost-effective office alternative to hydrocelectomy and spermatocelectomy.
- **Aspiration and sclerotherapy (AS)** remains a viable option for patients who are not surgical candidates or wish to avoid surgery.
- **Aspiration alone recurs in most cases** → a **sclerosing agent should always be added**.
- **Agents:** doxycycline, tetracycline, alcohol, polidocanol, ethanolamine oleate, sodium tetradecyl sulphate (STDS). Single-procedure success varies widely; the text names sterile alcohol, tetracycline and polidocanol as best.

TABLE 87.1 SCLEROSANTS WITH ASSOCIATED SINGLE-PROCEDURE AND RETREATMENT SUCCESS RATES

Class	Sclerosant	Single-procedure success (%)	Retreatment success (%)
Antibiotic	Tetracycline	57–85	57–90
Antibiotic	Doxycycline	84	88
Detergent	Polidocanol	54–67	83–89
Detergent	Sodium tetradecyl sulphate	62–76	94 upwards (upper bound not legible in source)
Alcohol-based	99.5% alcohol	76	98
Alcohol-based	5% ethanolamine oleate	68	98
Body text, all agents	—	33–85 (text range includes series not tabulated)	—

Sources as listed in the table: Jahnson 2011; Lund 2014; Sigurdsson 1994; Bullock and Thurston 1987; Daehlin 1997; Levine and DeWolf 1988; Shokeir 1994; Francis and Levine 2013; Beiko 2003; Braslis and Moss 1996; Khaniya 2009; Rencken 1990; Shan 2011; Tammela 1992.

- **What the table is for:** single-session success is modest with every agent; **retreatment is what raises success** to near-surgical levels.
- **Trap in the table:** polidocanol is named among the best agents, yet its single-procedure range is the lowest — its strength is on retreatment.

Aspiration and sclerotherapy technique.

- **Sequence:** prepare hemiscrotum with chlorhexidine → local anaesthetic at the aspiration site plus **cord block** (1% lidocaine with 0.25% bupivacaine) → **16–19-gauge angiocatheter** into the lateral or anterosuperior sac (plastic cannula preferred — less likely to slip out) → pre-connected via tubing to a **60-mL Luer-Lok syringe** (closed system) → drain completely and record volume → **microscopy for sperm**; culture or cytology if infection or malignancy is suspected → inject sclerosant into the collapsed sac and massage.
- **Recurrence** in the following weeks → repeat AS or proceed to hydrocelectomy.
- **Limitations:** pain, multiple treatments needed, poor efficacy in **multiloculated** and **large (>750 mL)** sacs.
- **AS versus hydrocelectomy:** AS has lower success and satisfaction but **fewer complications and lower cost**; repeat treatment brings success close to that of surgery. It is a safe, effective alternative for men wishing to avoid surgery.

Simple Orchiectomy

- **Bilateral simple orchidectomy** remains an option in hormone-sensitive metastatic prostate cancer for patients wishing to avoid medical androgen deprivation therapy (ADT).
- **1990s trend:** orchidectomy declined as gonadotrophin-releasing hormone (GnRH) agonist use rose.
- **Resurgence** is driven by the rising cost of medical castration, falling reimbursement and a reported survival advantage.
- **Evidence:** a retrospective comparison with luteinising hormone-releasing hormone (LHRH) agonists reported less prostate-specific antigen (PSA) rebound and better progression-free and overall survival after surgical castration; orchidectomy also caused less gynaecomastia and gave better quality of life (less physical discomfort and cancer-related anxiety). **A much larger later analysis did not confirm the survival benefit.**
- **Counsel** on the potential negative effect on self-image.
- **Preserving appearance:** testicular prosthesis or **epididymis-sparing (subepididymal) orchidectomy** — higher satisfaction than conventional orchidectomy, most patients satisfied cosmetically, and cost saving against medical castration.
- **Unilateral simple orchidectomy:** severe trauma, or prolonged ischaemia from torsion when salvage is impossible.
- **Chronic orchialgia** is a rare indication, **without level-one evidence** — only after all conservative treatment has failed.

Technique.

- **For orchialgia, the inguinal approach gives better pain outcomes**, probably because more proximal segments of the ilioinguinal, iliohypogastric and genitofemoral nerves are removed with the cord (Fig. 87.8); a small retrospective series found higher cure rates with inguinal than with scrotal orchidectomy.
- **Persistent pain after orchidectomy:** resection of the genital branch of the genitofemoral nerve gave excellent relief in most patients of a small series.
- **Scrotal orchidectomy sequence:** single vertical median raphe incision or bilateral transverse incisions → open all fascial and tunical layers → divide the cord into two bundles → double-clamp each proximally and distally → divide between clamps → **suture-ligate the proximal stump** with 2-0 Vicryl (polyglactin).
- **Epididymis-sparing sequence:** instead of dividing the cord, sharply dissect the testis off the epididymis under the **operating microscope** → find and suture-ligate the superior, middle and inferior epididymal arteries that perforate the testis → approximate the longitudinal epididymal edges with 4-0 chromic, ends left untied → tie the free ends to join caput and cauda.

Figure 87.8 — distribution of cutaneous nerves of the scrotum, perineum, thigh and groin (courtesy A. Lee Dellon).

- **Perineal map:** ilioinguinal; obturator; genitofemoral (genital and femoral branches); pudendal (dorsal, perineal and inferior rectal branches); posterior femoral cutaneous (perineal, thigh and inferior cluneal branches).
- **Thigh and groin map:** lateral femoral cutaneous, anterior femoral, iliohypogastric, ilioinguinal, genital and femoral branches of the genitofemoral, obturator and pudendal nerves.

Concept — *Surgical castration is regaining ground on cost and simplicity rather than on a proven survival advantage; for pain, the inguinal route works better because the pain is carried by nerves in the cord proximal to the testis, which a scrotal orchidectomy leaves behind.*

Testicular Prosthesis Placement

- **Consider for every male lacking a testis** (congenital, surgical, traumatic) — psychological distress from an empty scrotum occurs at any age; **offer before orchidectomy for testicular cancer.**
- **History:** Vitallium (1941), Lucite (1943) and glass (1953) prostheses.
- **Only Food and Drug Administration (FDA)-approved device: Coloplast Torosa** — silicone, four sizes (extra small to large), filled with saline through an injection port opposite the suture tab (Fig. 87.9).
- **Port rule:** pierce **no more than five times intraoperatively**; never access postoperatively.
- **Strict asepsis**, as for any prosthetic implantation.

COMPARISON OF PROSTHESIS INSERTION ROUTES

Route	Typical use	Risk profile
Inguinal	At radical orchidectomy; children with thin scrotal walls	—
High scrotal, into subdartos pouch	No malignancy, scrotum not compromised (surgeon preference)	Lower infection, erosion and pain — implant away from the suture line
Trans-scrotal, into subdartos pouch or tunica vaginalis	With simple orchidectomy for non-salvageable torsion or for ADT	Higher extrusion than high-scrotal or subinguinal — implant presses directly on the wound

Concept — *Once cancer control is settled, every choice of prosthesis route is about keeping the implant off its own suture line.*

EPIDIDYMIS

Excision of Epididymal Cysts and Spermatoceles

- **Origin:** cystic accumulations of semen in the epididymal tubules, rete testis or efferent ductules; mostly idiopathic, some after infective or traumatic inflammation.
- **Common benign ultrasound finding**, becoming more frequent with age.
- **Operate only for** discomfort, infection, impaired quality of life, disability from size, or infertility — as for hydrocele.
- **Counsel men of reproductive age** about **obstructive azoospermia**, particularly when the cyst lies in the corpus or cauda.
- **Sequence:** trans-scrotal incision → dissect to tunica vaginalis → sharply open tunica → dissect cyst down to its stalk → **ligate stalk with 5-0 or 6-0 absorbable suture**. If it cannot be freed entirely → unroof part of the wall → reapproximate the epididymal tunic with 4-0 absorbable suture.

Epididymal Tumour Excision

- **Possible malignancy** → **inguinal approach**, to avoid disrupting natural lymphatic drainage; cord clamped with non-crushing clamps pending frozen section.
- **Benign on preoperative percutaneous biopsy** → **trans-scrotal excision** is acceptable.
- **Microsurgical technique (10×–25× magnification)** to avoid injury to the testicular blood supply.

KEY POINTS: SURGERY OF THE SCROTAL CONTENTS

- High-resolution ultrasonography evaluates scrotal fluid collections; it may identify intratesticular malignancy and so guides the surgical approach.
- Excisional hydrocelectomy techniques have the lowest recurrence rates.

Concept — *Epididymal surgery is judged by its fertility cost: benign cysts are operated on only for symptoms because the price is obstructive azoospermia, and tumours follow the same inguinal-first oncological rule as the testis.*

VAS DEFERENS

Vasectomy

- **Safe, effective, permanent** male contraception; used by more than 10% of men in several countries (Australia, Bhutan, Canada, the Netherlands, New Zealand, the Republic of Korea, Great Britain, the United States).
- **Typical practice:** the great majority are performed on men aged 25–44, in the office, by urologists.
- **Sex gap:** in the United States, tubal ligation remains more prevalent than vasectomy (Centers for Disease Control and Prevention (CDC) National Survey of Family Growth, 2017–2019) — attributed to family-planning priorities, lower Medicaid coverage of vasectomy, and marital status (more unmarried women undergo tubal ligation than unmarried men undergo vasectomy).
- **Plan as for theatre surgery**, even in the office.
- **Counselling points: not immediately sterile** — other contraception until postvasectomy semen analysis (PVSA) confirms occlusion; **repeat vasectomy needed in <1%**; **pregnancy about 1 in 2000** even after an azoospermic PVSA, from recanalisation; **chronic scrotal pain about 1–2%**; alternatives; **signed consent**.
- **No prophylactic antibiotics** for routine vasectomy.
- **Anaesthesia:** local with or without oral sedation; intravenous sedation or general anaesthesia if local is declined or judged inadequate.
- **Local agents:** 1% or 2% lidocaine without adrenaline, alone or as a 50:50 mix with bupivacaine; long-acting liposomal bupivacaine (EXPAREL) gives good short- and long-term relief but may be cost-prohibitive.
- **AUA panel:** use the **smallest needle (25–32 gauge)** to reduce injection pain; **topical anaesthetic cream must not be the sole anaesthetic**.
- **Two key steps — isolation and occlusion.** Use the technique one is trained in, and change it if failures are significantly higher than expected.

Methods of Vas Isolation

COMPARISON OF VAS ISOLATION TECHNIQUES

Feature	Conventional vasectomy (CV)	Minimally invasive vasectomy (MIV)	No-scalpel vasectomy (NSV)
Skin opening	1.5–3 cm; single median raphe or bilateral paramedian	≤10 mm	Puncture, <10 mm
Dissection	Wide	Minimal	Minimal
Instruments	None special; towel clamp or Allis forceps	Vas-dissecting instruments and ring-tipped clamp	Ring-tipped clamp and sharp curved mosquito haemostat
Skin closure	Absorbable suture	May be left open	Absorbable suture or left open
Status	Reserved for difficult cases	Recommended	The original MIV technique

- **MIV is recommended** — less procedure-related discomfort and fewer complications (infection, bleeding, pain).
- **CV is still used** after prior scrotal surgery or trauma, with thick tight scrotal skin, with an associated varicocele, hydrocele or spermatocele, or whenever MIV isolation cannot be done safely.

Conventional Vasectomy

- **Single versus bilateral incision:** no study shows superiority; a single incision seems to cause fewer side effects and is quicker, whereas bilateral incisions avoid **grasping and dividing the same vas twice**. With a single incision, a gentle tug on the vas that moves the **ipsilateral testis** confirms the side.
- **Level of opening:** single midline incision at the penoscrotal junction or between it and the top of the testis; bilateral incisions at or above the penoscrotal junction to reach the **straight vas**. Lower openings reach the narrow **convoluted vas**, harder to handle now and at any later reversal.
- **Sequence: three-finger grasp** (thumb, index, middle) traps the vas; the same hand isolates both vasa, a few drops of fluid lubricating the skin → further anaesthetic precisely at the planned site → test with an Adson forceps (tugging and touch are expected, sharp pain is not; top up subcutaneously if needed) → longitudinal incision over the vasal sheath down to the vas → grasp with towel clamp or Allis forceps; overlapping clamps stop the vas slipping back → avoid adjacent structures, occlude only the vas → remove a short segment → occlude the free ends → check haemostasis → drop the ends back → repeat on the other side → close skin with absorbable suture (edges bleed and alarm patients).

Minimally Invasive Vasectomy

- **Definition:** any technique with a small (≤10 mm) skin opening and minimal vasal dissection, using special instruments — the sharp curved mosquito haemostat (Fig. 87.10) and the ring-tipped vas fixation clamp, whose cantilevered design prevents injury (Fig. 87.11).
- **Isolation:** the vas is raised to the anterior scrotal skin by the three-finger grasp.
- **Open versus closed access:** open — scalpel skin incision, minimal perivasal dissection, then clamp the vas; closed — clamp the vas through the skin first, transfer the clamp to the non-dominant hand, then incise. The remainder is common to all approaches.
- **Incisions may be left open** — they are small and bleed little.

No-Scalpel Vasectomy

- **Origin:** the first MIV technique, developed in China in 1974 by Li Shunqiang.
- **NSV is strictly a vas isolation technique** — it does not specify the occlusion method.
- **Sequence:** fix the vas in the ring-tipped clamp with the scrotal skin stretched tightly over it (Fig. 87.12) → after anaesthesia, puncture skin, sheath and vasal wall with the sharp curved mosquito haemostat (Fig. 87.13) → spread until the vas is reached → pierce the anterior vasal wall and lift the vas through the

opening (Fig. 87.14) → regrasp the partially delivered vas with the ring clamp → dissect and divide the posterior sheath to strip the vas clean, freeing the vasal artery (Fig. 87.15) → occlude → drop the ends back → close the puncture with absorbable suture or leave it open.

Figures 87.10–87.14 from Li, Goldstein, Zhu et al., *J Urol*, 1991.

Vas Occlusion

- **Success and failure depend directly on the occlusion method.**
- **United States practice:** complete division of the vas, with or without excision of a segment; ends separated by one or more methods below.

METHODS OF VAS OCCLUSION

Method	Description
Fascial interposition (FI)	Layer of internal spermatic fascia covers one vasal end; usually combined with another method
Suture ligation	Both ends; one to three ligatures about 1 cm from the cut edge
Clips	Replace sutures on the free ends
Folding back	Ends folded so they do not face each other — against recanalisation
Mucosal cautery (MC)	Thermal or electrical cautery of lumen and edge, a few millimetres up to 1.5 cm
Open-ended (partial ligation)	Abdominal end occluded, testicular end left open , FI between; limits pain from epididymal back-pressure; the sperm granuloma at the open end is linked to better reversal outcomes
Non-divisional extended cautery (Marie Stopes International)	No division; 2.5–3 cm electrocoagulation, anterior wall full-thickness, posterior wall partial-thickness; developed in the United Kingdom for low-resource settings; rare in the United States

AUA-RECOMMENDED OCCLUSION METHODS — OCCLUSIVE FAILURE RATE (OFR) ≤1%

Method	OFR
MC with FI, without clips or ligatures	0.0–0.55%
MC without FI and without ligatures or clips	0.0–0.60%
Open-ended: testicular end unoccluded; abdominal end MC + FI	0.0–0.50%
Non-divisional extended cautery	0.64%

- **The pattern:** every recommended method rests on **mucosal cautery**, alone or with FI — ligatures and clips are not the core of any of them.
- **Excision** of a segment is routine for most surgeons but not necessary; **if excised, 1.0 cm is adequate.**

Postvasectomy Semen Analysis

- **Test fresh semen at 8–16 weeks** (timing at the surgeon's discretion) after **at least 10–20 ejaculations**; older men clear sperm more slowly.
- **Persistent motile sperm** → repeat at **6 months** → still motile → **failure**; **arrange repeat vasectomy.**
- **Stop other contraception** when PVSA shows **azoospermia or rare non-motile sperm (RNMS, ≤100,000 non-motile sperm/mL)**. Stress this at counselling and again with postprocedure instructions.
- **Late recanalisation** remains possible: pregnancy after a successful vasectomy occurs at about **0.05%** — the same figure as 1 in 2000.

Vasectomy Reversal

- **Microsurgical reversal is the standard of practice** for men who wish to conceive naturally; covered in a separate chapter.

KEY POINTS: VASECTOMY

- No vasectomy technique is 100% effective.
- An initial in-person or telephone consultation — history, physical and genitourinary examination — precedes vasectomy.
- Success depends directly on the occlusion technique.
- Complete division of the vas, with or without excision of a segment, should be used; if excised, 1.0 cm is adequate.
- The open-ended technique may improve the chance of later reversal.
- Contraception may stop once PVSA shows azoospermia or RNMS ($\leq 100,000$ non-motile sperm/mL).

Concept — *Isolation and occlusion are separate decisions: isolation (CV, MIV, NSV) determines morbidity, occlusion determines efficacy, and only the PVSA proves success.*

SURGICAL MANAGEMENT OF CHRONIC SCROTAL PAIN

- **Pain** is a subjective sensation arising when noxious stimuli activate nociceptors that transmit from the peripheral to the central nervous system.
- **Chronic scrotal pain:** constant or intermittent pain **lasting more than 3 months** and interfering with daily activities; multifactorial, cause often undetermined; a common reason for clinic attendance, and patients typically consult several urologists.
- **Possible sources:** testis, epididymis, pampiniform plexus, post-vasectomy sperm extravasation, post-surgical haematoma, proximal injury to the ilioinguinal, iliohypogastric, genitofemoral, pudendal or sympathetic nerves.
- **Referred pain** from pelvic floor, prostate or other viscera sharing those nerve pathways.
- **Selection for surgery** rests on knowing the causative diseases, the scrotal nerve anatomy and a systematic approach.

FIGURE 87.16 ALGORITHM OF CHRONIC ORCHIALGIA EVALUATION AND MANAGEMENT

- **History, examination and duplex ultrasonography** → identifiable aetiology → treat the underlying condition.
- **No identifiable aetiology** → exclude an extrascrotal cause: computed tomography (CT) or magnetic resonance imaging (MRI) of abdomen and pelvis, spine or hip.
- **Referred from prostatitis or pelvic floor spasm** → treat prostatitis and/or start pelvic floor physiotherapy.
- **Non-specific pain localised to the scrotum** → trial of scrotal support, non-steroidal anti-inflammatory drugs (NSAIDs), tricyclic antidepressants and/or gabapentin → **pain persists beyond 3 months** → localise:
- **Localised to the epididymis** → epididymectomy.
- **Localised to testis/epididymis and/or began after vasectomy** → **diagnostic spermatic cord block** → **>50% improvement** → microsurgical spermatic cord denervation, or vasectomy reversal for post-vasectomy pain; **<50%** → pain clinic, with inguinal orchidectomy as the final surgical option.
- **Began after hernia repair** → **diagnostic ilioinguinal block** → **>90% improvement** → ilioinguinal neurolysis; **<90%** → pain clinic.

Modified from Tan and Levine, *World J Mens Health*, 2017.

Surgical Management of Intrasrotal Pain

Acute and Chronic Epididymitis

- **Acute epididymitis** usually resolves with rest, scrotal support, elevation, ice, antibiotics or NSAIDs.
- **Local softening (malacia)** of the epididymis suggests abscess; an abscess not responding to antibiotics is **drained or surgically debrided**.
- **Chronic epididymitis**: lasting **more than 6 weeks** despite adequate therapy.
- **Epididymectomy for epididymitis gives poor results — pain often persists → last resort**; pain-free rates are low and highly variable.
- **Indications for partial or total epididymectomy after failed medical therapy**: post-vasectomy epididymal engorgement, intractable epididymal pain, severe acute or chronic epididymitis, symptomatic epididymal cysts.
- **A palpable abnormality** (mass, cyst, granuloma) **significantly improves the chance of pain control**.
- **Counsel** on the low likelihood of complete resolution and the possible later need for orchidectomy.

Epididymectomy for Palpable Abnormalities of the Epididymis

- **Results are more favourable** than for epididymitis when there is a palpable abnormality such as a cyst.
- **Sequence** (cystic dilatation, granuloma, tender swollen epididymis): **ligate the efferent tubules** draining into the caput → sharply dissect corpus and cauda off the testis → in partial epididymectomy, **suture-ligate the proximal and distal stumps** to limit sperm extravasation.

Microsurgical Spermatic Cord Denervation

- **Indication**: non-specific chronic pain localising to the testis and/or epididymis, or post-vasectomy pain.
- **A spermatic cord block selects responders** to microsurgical denervation of the spermatic cord (MDSC), and **the degree of relief after block correlates with the degree of relief after MDSC**. In a prospective study, most patients were pain-free at 2 years.
- **Subinguinal sequence**: 4-cm incision over the external inguinal ring → isolate the cord → **divide the ilioinguinal nerve** (electrocautery) or ligate it (4-0 silk) → identify and ligate the **genital branch of the genitofemoral nerve** → operating microscope (4×–20×) with **micro-Doppler** to find arterial pulsation → **preserve the testicular and cremasteric arteries and the lymphatics** (the latter to prevent hydrocele) → divide the remaining cord structures — pampiniform veins, cremaster fibres and the perivascular tissue.
- **Preserve the vas and vasal vessels** while dividing the nerves in its sheath; the goal is to transect all genitofemoral branches (Fig. 87.18).
- **Scrotal alternative**: vertical paramedian scrotal incision to isolate the cord, with the same preservation of arteries and lymphatics.
- **Trifecta nerve complex** (Fig. 87.17) — a reproducible distribution of cord nerves, in decreasing density: **cremasteric muscle fibres → perivascular tissue and vasal sheath → posterior periarterial tissue/cord lipoma**. Immunohistochemistry confirms sensory and sympathetic nerves in the perivascular sheath and spermatic fascia.
- **Therefore, via the scrotal approach, the posterior periarterial tissue/cord lipoma must also be transected** to maximise pain control.

Figure 87.17 — cross-section of the right spermatic cord: cremasteric layer, vas, peri-cord lipoma, arteries, veins, lymphatics and biopsy sites (Parekattil et al., J Urol, 2013).

Varicocelectomy

- **Varicocele** (clinical or subclinical) is found in about **15% of men**.
- **Keep it in the differential of orchialgia**, especially a **dull ache worse on standing and exertion**.

- **Microsurgical repair gives more complete pain resolution than non-microsurgical repair** in men with palpable varicoceles and orchialgia.
- **Duration predicts success:** pain for more than 3 months made benefit from varicocelectomy more likely.

Retractile Testis and Intermittent Testicular Torsion

- **Postpubertal ascent** of one or both testes is rare but may cause pain and infertility.
- **Intermittent testicular torsion (ITT):** acute scrotal pain of **short duration (<2 hours)** with **rapid spontaneous resolution**; nausea and vomiting in about a quarter.
- **Orchidopexy is recommended for both** — low morbidity, excellent pain relief, reduced risk of testicular infarction.
- **Options:** suture fixation of the tunica albuginea, or placement in an extravaginal **dartos pouch**.

Technique.

- **Testicular fixation with suture:** midline or transverse incision → dissect to tunica vaginalis → open the parietal layer to expose testis and cord → fix tunica albuginea to dartos with **absorbable 3-0 suture**.
- **Dartos pouch:** transverse incision over the cranial hemiscrotum → dissect to tunica vaginalis → open parietal layer, externalise cord and testis → develop a **subdartos tunnel between dartos and external spermatic fascia** bluntly towards the dependent scrotum → place testis in the pouch → secure with a **purse-string around the cord** where it exits the cremaster window.

GOVERNING CONTRAST — SUTURE FIXATION VERSUS DARTOS POUCH

Suture fixation	Dartos pouch
Recurrent ipsilateral intravaginal torsion possible even with non-absorbable suture, occurring years after repair	No transparenchymal suture
Segmental or complete infarction if sutures entrap intratesticular arteries at the lower pole	—
Animal studies: transparenchymal chromic, nylon or catgut sutures → inflammation, intratesticular abscess, tubular necrosis, loss of spermatogonia	Animal studies: normal spermatogenesis in nearly all testes

- **If suturing, polytetrafluoroethylene (PTFE)** reduces the harm to spermatogenesis.
- **No consensus** on the ideal method; on animal data the authors recommend a **dartos pouch for patients wishing to preserve fertility**.

KEY POINTS: SURGERY OF THE SCROTAL CONTENTS

- Patients with chronic orchialgia have better pain outcomes when simple orchidectomy is performed through an inguinal approach.
- Men of reproductive age should be counselled on infertility risks before epididymal surgery.
- Epididymectomy for chronic epididymitis is a last-resort option because outcomes are poor and pain often persists.

Concept — *Surgery for chronic scrotal pain succeeds in proportion to how well the pain is localised: a palpable lesion, a positive cord block or a defined nerve territory predicts relief, whereas removing an organ for non-specific pain often fails because the pain generator lies proximally in the nerves.*

COMPLICATIONS OF SCROTAL SURGERY

Bleeding and Postoperative Haematoma

- **The main complication of scrotal surgery;** usually self-limiting.

- **Drain the haematoma** if substantial, infected or enlarging.
- **Continued bleeding** is most often from **incomplete ligation of the spermatic artery** → small subinguinal incision, retrieve the cord, re-ligate.
- **Retroperitoneal bleeding**: infrequent but **potentially lethal** — suspect with a falling haematocrit and an expanding haematoma; if the cord has retracted into the retroperitoneum, **extend the incision into the retroperitoneum**.

Chronic Scrotal Pain

- **Post-vasectomy pain significantly affecting quality of life: about 2%**; reports are of poor quality with variable definitions, and few men ultimately need surgery.

Early/Late Vasectomy Failure

- **Definition**: pregnancy, or failure to reach azoospermia or RNMS after vasectomy.

GOVERNING CONTRAST — EARLY VERSUS LATE VASECTOMY FAILURE

Early failure	Late failure
Technical — usually only one vas occluded properly	Recanalisation after a PVSA that met success criteria
Detected at PVSA	Motile sperm later and/or conception; about 1 in 2000

Epididymal/Vasal Injury or Ligation With Resulting Infertility

- **A known complication** of hydrocele and spermatocele surgery, but short- and long-term sequelae are little studied.
- **After AS**: sperm concentration, motility and morphology deteriorate at 6 months and **return to baseline by 12 months**.

Hydrocele Recurrence

- **0–6% after successful repair, regardless of operative technique.**
- **After AS**, resolution 85–96%, although repeat procedures may be needed to reach the higher rates.

Infection

- **Causes**: breach of sterile technique, or seeding of a seroma or haematoma.
- **Treatment**: antibiotics, or wound exploration, irrigation and haematoma evacuation; **a reopened incision is left open** for drainage and healing by secondary intention.

Sperm Granuloma

- **An inflammatory nodule at the vasectomy site is common and is not a complication.**
- **May be symptomatic for up to 3 months** → as-needed anti-inflammatory analgesia and reassurance.
- **Persisting symptomatic nodule**: <5%.
- **Prolonged pain localised to the vasal ends** → repeat vasectomy or excision of the vasal edges, with good outcomes.

Concept — *Scrotal complications trace back to two properties met in the anatomy — a space that cannot tamponade bleeding and ducts that obstruct if touched — so haematoma and iatrogenic obstruction are the complications to prevent, whereas a sperm granuloma is an expected response rather than a failure.*

ANATOMY AND EMBRYOLOGY OF SEMINAL VESICLES

- **First described** by Gabriele Fallopius in 1561; detailed study continued in the late 19th century.
- **Paired glands producing about 60% of the seminal fluid.**

- **Position:** behind the bladder base and above the prostate; each joins the **ampulla of the vas** to form the **ejaculatory duct**, which empties into the prostatic urethra at the **verumontanum** (Fig. 87.19).
- **Arterial supply mirrors the prostate:** inferior vesical, internal pudendal and middle rectal arteries.
- **Development sequence:** bilateral dorsolateral bulbous dilatations of the distal **mesonephric (Wolffian) ducts at about 12 weeks** → multiple diverticula by **16 weeks** → elongation and convolution → at birth the epithelium differentiates into **principal (secretory) and basal cells** → basal cells decrease until 18 years.
- **Size varies with age;** assessed by autopsy, transrectal ultrasonography (TRUS; Fig. 87.21), cross-sectional imaging (Figs. 87.22, 87.23) and prostatectomy specimens.
- **Normal adult seminal vesicle** (robotic prostatectomy specimens): about **3 cm long, about 7 mL** in volume; the **right is significantly more often larger than the left**.

Figures 87.20–87.23.

- **Fig. 87.20** (Langman, *Medical Embryology*, 1981): weeks 5, 8 and 13 — urogenital sinus, cloacal membrane, urorectal septum, mesonephric duct and ureteral bud; by week 13 the SV buds from the vas beside the developing prostate.
- **Fig. 87.21:** TRUS of normal SVs, transverse and sagittal (bladder, ejaculatory duct, urethra labelled). **Fig. 87.22:** CT of an SV cyst beside the bladder and ureters. **Fig. 87.23:** transaxial MRI of normal SVs with endorectal coil, T1- and T2-weighted.

Concept — *The seminal vesicle is a Wolffian derivative sitting on the ejaculatory pathway, which is why its cysts travel with Wolffian and renal anomalies and why its disease so often presents as ejaculatory duct obstruction.*

SURGICAL APPROACHES TO SEMINAL VESICLES

- **Most SV procedures are performed with radical cystectomy or prostatectomy;** approaches to unilateral or bilateral SV surgery therefore mirror those for pelvic neoplasms.

COMPARISON OF APPROACHES TO THE SEMINAL VESICLES

Approach	Indication	Defining feature	Principal risk
Anterior — transvesical transtrigonal	Large benign SV masses or cysts; ectopic ureter draining into an SV cyst	Most direct; bladder opened anteriorly and posteriorly	Urinary leak; ureteric injury
Anterior — perivesical	Children with a large SV cyst, especially if nephroureterectomy is planned	Extravesical; bladder mobilised off the lateral wall	Ureter (superior) and vas (superomedial)
Anterior — transperitoneal (open, laparoscopic, robotic)	Bilateral SV cysts or benign masses	Rectovesical plane entered through the cul-de-sac; now dominant	Laparoscopic access injuries
Perineal	At perineal radical prostatectomy	Small exposure	Neurovascular bundle injury; incomplete SV removal
Posterior — transcoccygeal	Previous suprapubic or perineal surgery; cannot tolerate supine or lithotomy	Prone jackknife; least used	Rectal injury

Anterior

- **Transvesical transtrigonal sequence** (Fig. 87.24): infraumbilical midline incision → space of Retzius → vertical anterior cystotomy → self-retaining retractor exposes posterior wall → **bilateral ureteric catheters or feeding tubes** to identify and protect the ureters → trigonal incision avoiding the bladder neck (Fig.

87.24: vertical between the orifices, or transverse 2 cm above the bladder neck below the orifices) → stay sutures on the cut edges → through the detrusor, the ampullae appear below bladder-neck level → **medial traction on the vas identifies the ipsilateral SV** → excise the SV at its junction with the vas, or together with the ampulla → **two-layer posterior closure** (2-0 or 3-0 Vicryl to muscularis, 4-0 Vicryl to mucosa) → close anterior wall likewise → closed suction pelvic drain and Foley catheter. **Ureteric stents only if injury is suspected** — not standard in uncomplicated cases.

- **Perivesical sequence:** midline or Pfannenstiel incision → dissect the bladder off the lateral sidewall on the cyst side → dissect the SV down to its junction with the prostate → ligate with 2-0 absorbable suture and a clip, avoiding the **ureter superiorly** and the **vas, which runs superior and medial** → excise → standard closure; drain and Foley for 24 hours, removed if there was no bladder injury and drain output is low.
- **Transperitoneal sequence** (open, laparoscopic or robotic): enter pelvis → transverse incision in the peritoneum where it meets the bladder in the cul-de-sac → develop the space between bladder and rectum (Fig. 87.25) → identify SVs and ampullae on the posterior bladder → dissect, ligate, resect.
- **History:** laparoscopic SV dissection was first described by Kavoussi (1993) during laparoscopic pelvic lymph node dissection before perineal prostatectomy — SVs and ampullae freed while awaiting frozen section; if nodes were negative, the patient was repositioned for perineal prostatectomy. It was adapted to the robotic approach (Fig. 87.26), now the route for most pelvic urological surgery.

Figures 87.24, 87.25 and 87.27 are redrawn from Hinman, *Atlas of Urologic Surgery*, 1989; in Fig. 87.25 the ampullae lie in the midline with the SVs immediately lateral, and the SV duct is ligated and transected. Figure 87.26 — positioning for laparoscopic SV dissection; **diamond port configuration**, and **inverted-U configuration in obese patients** (Winfield, 1993).

Perineal

- **Mostly used at perineal radical prostatectomy.**
- **Limited incision** makes dissection of the ampullae and SVs difficult; poor visibility risks **neurovascular bundle injury or incomplete SV removal**.
- **Stay as close to the midline as possible** to protect the bundles; laparoscopic SV dissection has been described to facilitate this route.

Posterior

- **Transcoccygeal approach** (Fig. 87.27): least used and least familiar, but useful after suprapubic or perineal surgery, or when supine or lithotomy positioning is not tolerated.
- **Sequence: prone jackknife** → **L-shaped incision** vertically from mid-sacrum, turning at the coccygeal tip towards the gluteal cleft, 3 cm from the rectum → dissect to the lateral side of the coccyx → retract gluteus maximus fibres → dissect the rectosigmoid off the sacrum → mobilise the ipsilateral lateral rectal wall medially until the prostate is reached (Denonvilliers fascia incised, Fig. 87.27C) → superomedial dissection reveals the ampulla in the midline with the SV lateral to it → Penrose drain through a separate stab incision, removed at 2–3 days if dry → inspect the rectum; **imbricate in two layers if injured** → return rectum to position, close in layers.

Concept — *Choosing an SV approach trades directness against collateral risk: anterior routes endanger bladder and ureter, perineal and posterior routes endanger the neurovascular bundles and rectum, and the robotic transperitoneal route has displaced the others because it opens the rectovesical plane without opening either organ.*

SURGERIES OF THE SEMINAL VESICLES

Seminal Vesicle Cyst Management

- **Rare:** prevalence about **0.005%** of men.

GOVERNING CONTRAST — CONGENITAL VERSUS ACQUIRED SEMINAL VESICLE CYST

Congenital	Acquired
Mesonephric (Wolffian) duct abnormalities	Ejaculatory duct obstruction secondary to infection, urethral inflammation or ejaculatory duct stones

- **Most are asymptomatic.** Symptoms: haematospermia, terminal haematuria, lower abdominal or lumbosacral pain, testicular and perineal discomfort; obstructive hydronephrosis; voiding or defecatory problems.
- **Zinner syndrome:** SV cyst + ipsilateral renal agenesis + ejaculatory duct obstruction (often with an ectopic ureter).
- **Management by size and symptoms:** small, asymptomatic or mildly symptomatic → conservative, occasionally percutaneous drainage or transrectal or transurethral aspiration; **large with significant symptoms → surgical excision.**
- **First excision** of a symptomatic SV cyst: Zinner, 1914, who also described the associated anomalies. Today excision is **laparoscopic or robotic** — lower morbidity, shorter stay, more accurate dissection.

Transurethral Endoscopic Treatments

- **Preferred approach** to SV cysts, persistent haematospermia, and ejaculatory duct obstruction and/or stones — less invasive, faster, fewer complications.
- **Transurethral unroofing** with a standard resectoscope: resect an area about **1 cm long × 1 cm wide × 1 cm deep** between bladder neck and verumontanum → once unroofed, rinse the cyst with saline and cauterise with the loop; easy access, little trauma or bleeding, few complications, rapid recovery.
- **Transurethral seminal vesiculoscopy:** semirigid ureterscope after serial ureteric dilatation; the **holmium laser** is safe and effective for incising obstructed ducts, coagulating haemorrhagic mucosa and fragmenting duct or SV stones.
- **In infertility:** sperm appeared initially in most men but fell after 6 months — re-obstruction is common, from the underlying disease or the procedure.

Transrectal Ultrasound-Guided Aspiration

- **TRUS suggests obstruction** when the **SV width is >1.5 cm** and the **ejaculatory duct diameter >2.3 mm**.
- **Interpret cautiously:** a dilated SV does not prove obstruction, and obstructed ducts may have undilated SVs.
- **Confirm by SV aspiration showing sperm.**

Seminal Vesicle Tumour Excision

- **Primary SV tumours are extremely rare;** benign and malignant both appear as complex solid-cystic retrovesical masses, often asymptomatic, with symptoms overlapping cysts, stones, inflammation and metastases.
- **Work-up:** abdominal and pelvic imaging first; **MRI is the most accurate tool** (multiplanar, excellent tissue contrast); **TRUS-guided biopsy** if imaging is inconclusive; establish benign versus malignant preoperatively because malignancy may need radical surgery.

PRIMARY SEMINAL VESICLE TUMOURS

Benign	Malignant
Cystadenoma	Mixed epithelial–stromal tumour (MEST)
Hydatid cyst	Phyllodes tumour
Papillary adenoma	Adenocarcinoma
Amyloid deposition	Sarcoma; carcinoid; primary seminoma; yolk sac tumour

- **Resection:** open or laparoscopic; open approaches are being replaced by laparoscopic and robotic ones (fewer complications, shorter stay). The commonest open route is **transvesical**, which exposes the SVs without damaging the pelvic plexus or the blood supply of bladder and rectum. **Robotic nerve-sparing vesiculectomy** gives safe perioperative and oncological outcomes.
- **Primary SV adenocarcinoma:** about 60 reported cases; **no standard guidelines**. Multimodal treatment with **radical excision as the mainstay**. Most present late with local spread, mainly to bladder and prostate → radical prostatectomy, radical cystoprostatectomy, partial cystectomy or total pelvic exenteration (ileal conduit and sigmoid colostomy), with bilateral lymphadenectomy and adjuvant hormonal, radiation or chemotherapy.
- **Yolk sac tumour of the SV:** highly malignant; only two case reports; presents with haematospermia; multimodal chemotherapy, radiotherapy and radical surgery described but not a standard of care.

KEY POINTS: SURGERY OF THE SEMINAL VESICLES

- SVs produce about 60% of the ejaculate fluid.
- Surgical approaches to the SVs mirror those for pelvic neoplasms.
- Symptomatic SV cysts may be associated with ipsilateral renal agenesis and ectopic ureter.
- TRUS-guided SV aspiration must be performed to confirm ejaculatory duct obstruction, because the SV may be dilated for a non-obstructive reason.
- Benign and malignant SV tumours are very rare.
- MRI provides the most accurate radiographic evaluation of SV lesions.

Concept — Seminal vesicle cysts are treated from the urethra because most are the upstream consequence of ejaculatory duct obstruction: opening the duct treats the cause, and excision is kept for large symptomatic cysts and tumours.

COMPLICATIONS OF SEMINAL VESICLE SURGERY

- **SV surgery mimics prostate surgery** and carries similar complications.

COMPLICATIONS OF SEMINAL VESICLE SURGERY BY NEIGHBOURING STRUCTURE

Injury	Why at risk	Management
Bladder	Anterior dissection; transvesical route opens the bladder twice (anterior and posterior)	Two-layer closure (mucosa, muscle) + prolonged drainage; transvesical: prolonged catheterisation, consider pelvic drain
Rectum	Directly behind the SVs	No gross faecal spillage → debride devitalised tissue, two-layer primary repair ; extensive injury, gross spillage or doubtful repair → add temporary diverting colostomy ; risk minimised by an anterior approach
Neurovascular bundles	Lie immediately lateral to the SV tips	Cautious dissection; injury → transient or permanent erectile dysfunction

Injury	Why at risk	Management
Ureter	Enters the posterior bladder anterolateral to the SV tips , crossing towards the midline above the trigone	Confirm with intravenous indigo carmine or on-table retrograde ureterogram → ureteroneocystostomy (pull ureter into bladder, spatulate, suture to mucosa); tension → psoas hitch or Boari flap; stent

- **Endoscopic complications:** epididymitis and post-void dribbling from urinary reflux, and urinary tract infection.
- **Laparoscopic/robotic complications:** uncommon but potentially devastating, possibly forcing termination or open conversion — trocar injury to vessels or bowel, abdominal wall bleeding, extraperitoneal insufflation, gas embolism.

Concept — *Every complication maps to a neighbour of the seminal vesicle — bladder in front, rectum behind, neurovascular bundles laterally, ureter above — so the approach chosen decides which neighbour is placed at risk.*